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New isoindolinones from the fruiting bodies of the fungus *Hericium erinaceus*

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ABSTRACT

Two new isoindolin-1-ones, named erinacerin (M–N) along with eleven known compounds have been isolated from the fruiting bodies of the medicinal mushroom *Hericium erinaceus*. The structures of compounds were elucidated based on spectroscopic studies and comparison with the known related ones. The anticancer potential of the isolates were tested with an *in vitro* cytotoxic assay against six human cancer cell lines (SH-SY5Y, 1321N1, HCT-116, Caco-2, OVK18 and HeLa).

1. Introduction

Hericium erinaceus is a well-known edible and medicinal mushroom. It has been used as a traditional Chinese medicinal fungus for many years to treat gastrointestinal diseases as dyspepsia and gastric ulcers (Wang et al., 2016). The mushroom is called Yamabushitake in Japanese, Houtougu (monkey head) in Chinese, and Lion's Mane in English after its shape (Kim et al., 2011). It has been demonstrated that *H. erinaceus* contains many bio-active compounds, which have shown various biological activities including anticancer (Mizuno et al., 1992), prevention and treatment of Alzheimer's disease (FURUTA et al., 2016), antibacterial, anti-inflammatory (Kim et al., 2012), antioxidant (Zhang et al., 2006), anti-obesity (Hiraki et al., 2017) and stimulating the synthesis of nerve growth factor (Kawagishi et al., 1994, 1991). During the past two decades, the chemical constituents *H. erinaceus* have been investigated, which includes polysaccharides (HOU et al., 2015; Jia et al., 2004; Wang et al., 2015b; Wu et al., 2015a), sterols (Li et al., 2014; Takaishi et al., 1991), fatty acids (KAWAGISHI et al., 1990; Miyazawa et al., 2008), hericerins (Li et al., 2015), erinacines (Tang et al., 2015), erinacerins (Wang et al., 2015a) and erinaceolactones (Wu et al., 2015b). In the course of our continuing search for new active compounds from mushrooms (Amen et al., 2016b, a; Satria et al., 2018; Shimizu et al., 2016), two new aromatic compounds, namely erinacerin M and N (Fig. 1) were isolated. In this report, we describe the isolation and structural elucidation of these new compounds. We describe further, the cytotoxicity of the isolated compounds against several cancer cell lines.

2. Results and discussion

Chemical studies of the different fractions of *H. erinaceus* using different chromatographic techniques afforded 13 compounds (**1–13**), of which 11 were identified as stearic acid (**3**), oleic acid (**4**), hericenone D (**5**), hericenone I (**6**), hericene A (**7**), ergosteryl stearate (**8**), ergosterol (**9**), ergosterol peroxide (**10**), erinaceolactam C (**11**), nicotinamide (**12**) and adensoine (**13**).

Compound 1 was obtained as a white amorphous powder, $[\alpha]_D^{20}$ +2.0 (c 0.01, MeOH). HR-FAB-MS spectrum showed a protonated molecular ion peak at m/z 344.1862 $[M+H]^+$, calculated 344.1854, in accordance with a molecular formula $C_{20}H_{25}NO_4$. It exhibited UV absorption at 288 nm. A detailed analysis of NMR data in comparison with those in literature suggested that compound **1** has a partial structure of erinacerin A (Yaoita et al., 2005). ^{13}C-NMR spectrum (Table 1) revealed the presence of twenty signals. The most indicative signals are the carbonyl carbon at δ_C 198.1, the olefinic carbons at δ_C 155.6 and δ_C 125.1 indicating conjugation of double bond with carbonyl group, carbonyl of γ-lactam at δ_C 169.1 (Yaoita et al., 2005), one methylene carbon linked to γ-lactam nitrogen at δ_C 49.6 and an oxygenated quaternary carbon at δ_C 75.9 as well as a methoxyl functionality at δ_C 55.9.

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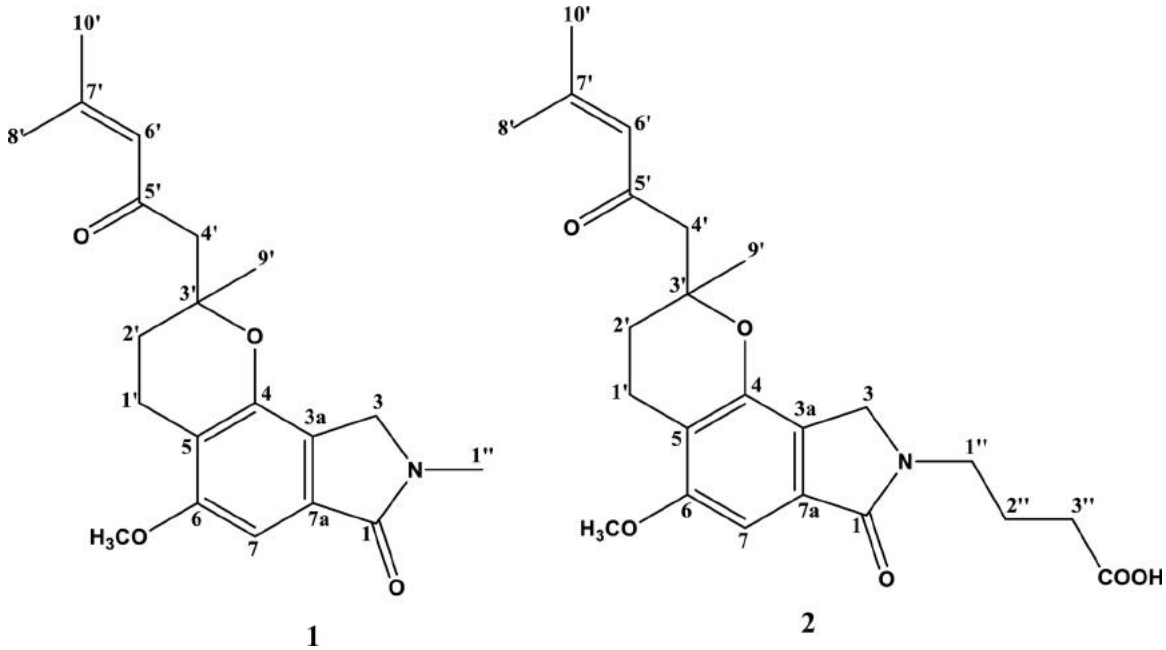

Fig. 1. Structures of compounds 1 and 2 from *Hericium erinaceus.*

Table 1
^{1}H and ^{13}C-NMR spectra data of compounds 1 and 2*.

	1		2	
#	^{1}H (*J HZ*)	^{13}C	^{1}H (*J HZ*)	^{13}C
1		169.1		171.2
3	4.18 (d, 16.8), 4.21 (d, 16.8)	49.6	4.27 (d, 17.4), 4.32 (d, 17.4)	48.0
3a		132.3		132.9
4		148.6		150.1
5		113.5		115.3
6		158.5		160.2
7	6.89 (s)	96.2	6.87 (s)	96.8
7a		121.7		123.8
1'	2.69 (t, 6.6)	19.1	2.72 (t, 6.6)	18.4
2'	1.92 (m), 2.01 (m)	30.7	1.92 (m), 2.02 (m)	31.9
3'		75.9		77.3
4'	2.63 (d, 13.8), 2.76 (d, 13.8)	52.2	2.62 (d, 13.8), 2.82 (d, 13.8)	53.0
5'		198.1		200.4
6'	6.06 (brs)	125.1	6.15 (brs)	126.3
7'		155.6		157.4
8'	1.86 (d, 0.6)	27.8	1.88 (d, 0.6)	27.9
9'	1.41 (s)	24.8	1.40 (s)	25.2
10'	2.15 (d, 0.6)	20.8	2.13 (d, 0.6)	20.9
6-OCH_3	3.87 (s)	55.9	3.88 (s)	56.4
1"	3.17 (s)	29.7	3.63 (t, 7.2)	43.6
2"			1.95 (m)	26.2
3"			2.24 (t, 7.2)	35.3
3"-COOH				180.9

*Chemical shifts (δ) are expressed in ppm and coupling constants (*J*) in Hz. ^{1}H and ^{13}C NMR were measured in $CDCl_3$ (for **1**) and CD_3OD (for **2**) at 600 and 150 MHz, respectively.

There were nine degrees of unsaturation in the molecule according to the molecular formula. One aromatic ring, γ-lactam and α,β-unsaturated ketone accounted for eight of them. One degree of unsaturation remained suggesting the presence of one more ring in the structure. ^{1}H-NMR data (Table 1) showed signals due to tertiary methyl groups [δ_H 1.41 (3H, CH_3-9')], two olefinic methyl groups [δ_H 1.86 (3H, CH_3-8') and δ_H 2.15 (3H, CH_3-10')]. It also showed one singlet aromatic proton at δ_H 6.89 suggested the presence of penta-substituted aromatic ring. The proton δ_H 6.89 showed cross peaks in HMBC spectrum with carbon signals at δ_C 113.5 (C-5), δ_C 132.3 (C-3a), δ_C 121.7 (C-7a), δ_C 148.6 (C-4) and δ_C 158.5 (C-6). Furthermore, the peaks at δ_H 4.18 and 4.21, assigned for H-3 showed HMBC correlations with the carbon signals at δ_C 132.3 (C-3a), δ_C 121.7 (C-7a), δ_C 148.6 (C-4) and δ_C 169.1 (C-1) confirming the attachment of aromatic ring to γ-lactam ring. A second singlet proton signal at δ_H 6.06, assigned to H-6', showed cross peaks with the carbon signals at δ_C 20.8 (C-10') and δ_C 155.6 (C-7') together with the correlations of δ_H 2.63 (d, $J = 13.8$ Hz), δ_H 2.76 (d, $J = 13.8$ Hz) assigned for H-4' with δ_C 125.1 (C-6') and δ_C 198.1 (C-5') confirming the presence of α,β-unsaturated ketone. Also H-4' showed cross peaks with the carbon signals at δ_C 30.7 (C-2') and δ_C 24.8 (C-9') and δ_C 75.9 (C-3'). The correlation between H-4' and carbon signal at δ_C 75.9 furthermore confirmed the presence of an extra oxygenated ring in the structure. On the basis of the above discussion, and by comparison of these spectroscopic data with those of erinacerin A (Yaoita et al., 2005), the skeleton of **1** should be the same except for the absence of the side chain attached to nitrogen atom and instead the presence of a methyl group evidenced by significant HMBC correlations observed between the singlet peak at δ_H 3.17 assigned for H-1" with δ_C 121.7 (C-7a) and δ_C 169.1 (C-1). Consequently, the structure of **1** was established as a new natural product, named erinacerin M. Compound **1** has an asymmetric carbon. Its circular dichroism (CD) spectrum exhibited a negative cotton effect at 209 nm [Δε -1.30]. Its ^{1}H and ^{13}C-NMR are shown in Table 1 and Its key HMBC correlations are shown in Fig. 2. Erinacerin A-L (Wang et al., 2015a; Yaoita et al., 2005) have been isolated before so the new compound was named erinacerin M to complete this series of interesting natural products.

Compound 2 was obtained as a white amorphous powder, $[\alpha]_D^{20}$ +3.0 (c 0.01, MeOH). HR-FAB-MS spectrum showed a protonated molecular ion peak at m/z 416.2071 $[M+H]^+$, calculated 416.2065, in accordance with a molecular formula $C_{23}H_{29}NO_6$. It exhibited UV absorption at 288 nm. Careful investigation of the spectroscopic data revealed that it has the same skeleton as erinacerin M except for the presence of a propionic acid moiety attached to nitrogen atom instead of a methyl group evidenced by significant HMBC correlations observed between δ_H 3.63 (t, $J = 7.2$ Hz) assigned for H-1" with δ_C 26.2 (C-2"), δ_C 35.3 (C-3"), δ_C 48.0 (C-3) and δ_C 171.2 (C-1) together with correlations observed between δ_H 1.95 (m), assigned for H-2" with δ_C 35.3 (C-3"), δ_C 43.6 (C-1") and δ_C 180.9, assigned for a carboxylic moiety and further

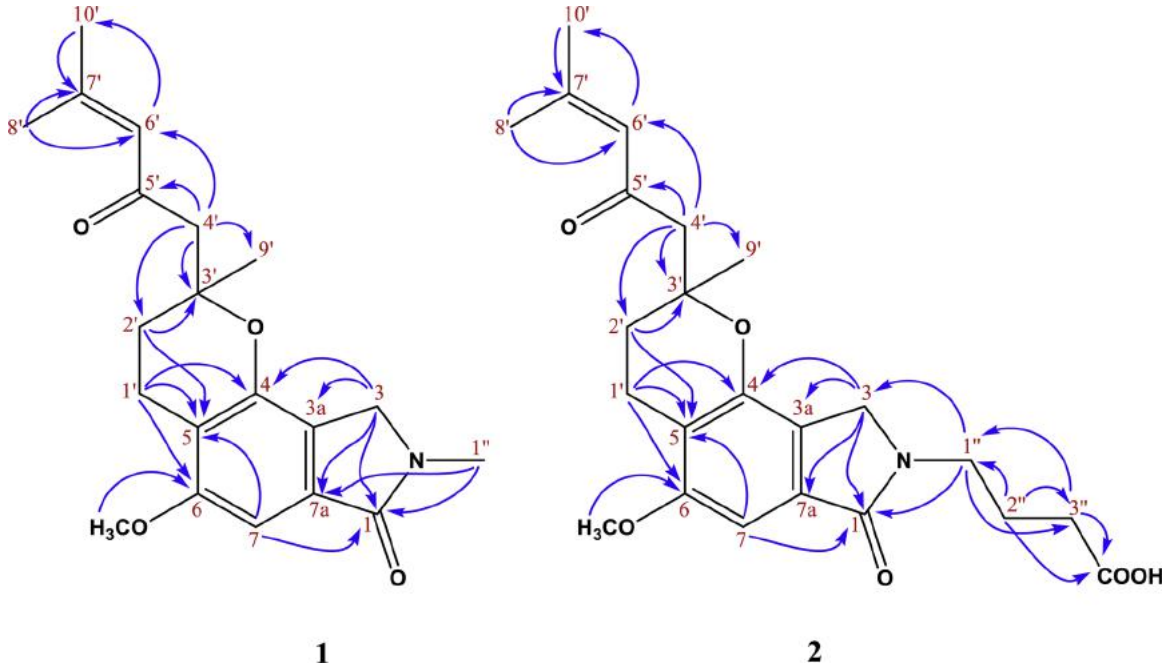

Fig. 2. Key HMBC correlations of compounds 1 and 2.

confirmed through correlations of δ_H 2.24 (t, $J = 7.2$ Hz), assigned for H-3" with δ_C 43.6 (C-1") and δ_C 180.9 (3"-COOH − COOH). Consequently, the structure of **2** was also established as a new natural product, named erinacerin N. Same as **1**, compound **2** has a symmetric carbon and its circular dichroism (CD) spectrum exhibited a negative cotton effect at 213 nm [Δε -2.0]. Its ^{1}H and ^{13}C-NMR are shown in Table 1 and Its key HMBC correlations are shown in Fig. 2.

In addition to the three compounds described above, eleven known compounds were identified by comparing their physicochemical and spectral data with those in the literature. These compounds were identified as stearic acid **(3)**, oleic acid **(4)**, hericenone D **(5)** (Kawagishi et al., 1991), hericenone I **(6)** (Ueda et al., 2008), hericene A **(7)** (Miyazawa et al., 2012), ergosteryl stearate **(8)** (Li et al., 2014), ergosterol (**9**), ergosterol peroxide (**10**) (Yusoo et al., 2001), erinaceolactam C (**11**) (Wang et al., 2016), nicotinamide (**12**) (Yamashita et al., 1989) and adenosine (**13**) (Kiem et al., 2008).

The cytotoxicity of the isolated compounds was evaluated against six human cancer cell lines; human ovarian endometrioid carcinoma cells (OVK18), human neuroblastoma cells (SH-SY5Y), human brain carcinoma (astrocytoma) cells (1321N1), human colorectal carcinoma cells (Caco-2, HCT-116) and human cervical carcinoma cells (HeLa). The results are summarized in Table 2. Compounds **6** and **10** showed a strong cytotoxic activity against all the tested cell lines. Compound **8** was found to have strong activity against the cancer cell lines SH-SY5Y, HCT-116 and OVK18.

3. Experimental

3.1. General

Optical rotations were measured with a Jasco DIP-370 digital polarimeter at 20 °C. Ultraviolet (UV) spectra were obtained using UV–vis spectrophotometer (Shimadzu 1601 PC, model TCC240, Kyoto, Japan). ^{1}H and ^{13}C-NMR spectra were obtained on a Bruker DRX 600 NMR spectrometer (Bruker Daltonics Inc., MA, USA) using TMS as an internal standard for chemical shifts. Chemical shifts (δ) were expressed in ppm with reference to the TMS resonance. HR-FAB-MS were measured with a JEOL JMS 700 spectrometer (JEOL, Japan). CD spectra were measured with a Jasco J-720 W spectropolarimeter at 22 °C. Organic solvents were purchased from Wako Pure Chemical Industries (Osaka, Japan). Silica gel (75–120 mesh) and RP-C_{18} silica (38-63 μm) were purchased from Wako Pure Chemical Industries (Osaka, Japan). Diaion HP-20 was purchased from Mitsubishi Chemical Co. (Tokyo, Japan). Thin layer chromatography (TLC) silica gel 60 F_{254} plates were purchased from Merck Co., Darmstadt, Germany. Plates were developed in a solvent mixture of different organic solvents, and the developed chromatograms were visualized under 254 nm UV light and the spots were made visible by spraying with vanillin/H_2SO_4 reagent before warming in an oven preheated to 110 °C for 5 min. Preparative TLC (PTLC) was performed on precoated RP-C_{18} F_{254} plates (5 x 7.5 cm x 0.2 mm thick) on glass plates.

3.2. Fungal material

The 50% ethanolic extract used in this study was provided by Aso Biotech Co., Ltd. (Kumamoto, Japan).

3.3. Extraction and isolation procedures

The 50% ethanolic extract solution (1.0 kg) was exhaustively partitioned successively with *n*-hexane and EtOAc. The different fractions were evaporated under reduced pressure at 40 °C to obtain *n*-hexane fraction (fraction A, 5 g), and EtOAc fraction (fraction B, 13 g). **Fraction A** was applied onto the top of a silica gel column (30 x 4 cm) previously packed in *n*-hexane. Gradient elution was performed using *n*-hexane containing increasing proportions of EtOAc (100:0→ 0:100) followed by EtOAc with increasing proportions of MeOH (100:0→ 80:20). The effluents were collected in 50 mL fractions. Each fraction was concentrated to a small volume under reduced pressure at 40 °C and monitored by TLC. Fractions with the same chromatographic pattern were pooled together and evaporated to dryness. Fractions 19–35 [eluted with *n*-hexane–EtOAc (97:3)] was purified over a silica gel column using *n*-hexane–DCM (100:0→80:20) to isolate compounds **3** (8 mg) and **4** (13 mg). Fractions 46–68 [eluted with *n*-hexane–EtOAc (95:5)] was applied onto the top of RP-C_{18} column using a gradient elution of H_2O–MeOH (98:2→10:90) to give compounds **5** (10 mg), **6** (6 mg) and **7** (1.8 mg). Fractions 73–90 [eluted with *n*-hexane–EtOAc (90:10)] was chromatographed over a silica gel column using a gradient

Table 2
Cytotoxic activities against SH-SY5Y, 1321N1, HCT-116, Caco-2, OVK18 and HeLa*.

Compounds	IC_{50} (μM)					
	SH-SY5Y	1321N1	HCT-116	Caco-2	OVK18	HeLa
1	> 50	> 50	> 50	> 50	> 50	> 50
2	> 50	> 50	> 50	> 50	> 50	> 50
3	> 50	> 50	> 50	> 50	> 50	> 50
4	> 50	> 50	> 50	> 50	> 50	> 50
5	> 50	> 50	> 50	> 50	> 50	> 50
6	36.69 ± 2.44	41.66 ± 3.20	7.66 ± 0.87	49.53 ± 2.00	0.99 ± 0.78	25.94 ± 9.23
7	> 50	> 50	31.06 ± 6.58	> 50	> 50	> 50
8	35.52 ± 3.21	> 50	2.77 ± 0.32	> 50	8.1 ± 1.03	> 50
9	> 50	> 50	> 50	> 50	> 50	> 50
10	8.84 ± 2.02	21.68 ± 0.22	52.73 ± 4.35	6.35 ± 2.75	8.07 ± 0.41	33.04 ± 2.74
11	> 50	> 50	> 50	> 50	> 50	> 50
12	> 50	> 50	> 50	> 50	> 50	> 50
13	> 50	> 50	> 50	> 50	> 50	> 50
5-FU	> 50	> 50	> 50	> 50	29.21 ± 7.17	> 50

* The activity was shown as IC_{50} value, which was the concentration of the tested compound (μM) that decreased the number of viable cells by 50%. Results are expressed as IC_{50} ± SD (n = 3).

elution of *n*-hexane–EtOAc (100:0→85:15) to give compound **8** (12 mg). Fractions 91–112 [eluted with *n*-hexane–EtOAc (85:15)] was crystallized from acetone several times under cooling to give compound **9** (40 mg). Fractions 113–146 [eluted with *n*-hexane–EtOAc (75:25)] was chromatographed over a silica gel column using a gradient elution of *n*-hexane–acetone (100:5→ 80:20) to yield compound **10** (20 mg). **Fraction B** was applied onto the top of a silica gel column (42 x 6 cm) previously packed in *n*-hexane. Gradient elution was performed using *n*-hexane containing increasing proportions of EtOAc (10:90→0:100) followed by EtOAc with increasing proportions of MeOH (100:0→ 0:100). The effluents were collected in 150 mL fractions. Each fraction was concentrated to a small volume under reduced pressure at 40 °C and monitored by TLC. Fractions with the same chromatographic pattern were pooled together and evaporated to dryness. Fractions 59–76 [eluted with EtOAc–MeOH (95:5)] was applied onto the top of RP-C_{18} column using a gradient elution of H_2O–ACN (100:0→20:80) to give compounds **11** (10 mg), **1** (2 mg) and **2** (3.6 mg). PTLC (on precoated RP-C18 F_{254} plates) was performed for subfractions 81–90 [eluted with EtOAc–MeOH (90:10)] using MeOH–H_2O (50:50) to yield compound **12** (2.3 mg). Fraction 92–113 [eluted with EtOAc–MeOH (90:10)] was chromatographed on Diaion HP-20, eluted with water then with methanol/water to give compound **13** (8.5 mg), eluted with H_2O–MeOH (90:10).

3.4. Cell culture

All cell lines were obtained from Riken Bio resource Center of Japan (Ibaraki, Japan). The medium for culturing human neuroblastoma cells (SH-SY5Y), human brain carcinoma (astrocytoma) cells (1321N1), human colorectal carcinoma cells (Caco-2) and human cervical carcinoma cells (HeLa) is Dulbecco's Modified Eagle's medium (DMEM) (Wako Pure Chemical Industries, Osaka, Japan). Human colorectal carcinoma cells (HCT-116) were cultured in McCoy's 5 A modified medium (Gibco BRL, Tokyo, Japan). The medium for culturing human ovarian endometrioid carcinoma cells (OVK18) is Eagle's minimal essential medium (EMEM) (Wako Pure Chemical Industries, Osaka, Japan). All culture media were supplemented with 10% fetal bovine serum (FBS) (Hyclone, Logan, UT, USA), 100 U/mL penicillin and 100 U/mL streptomycin (Wako Pure Chemical Industries, Osaka, Japan). Cytotoxicity was determined as follows: cells were cultured in 96-well plates at densities of 0.5×10^4 cells/well, in a humid atmosphere of 5% (v/v) CO_2 and 95% (v/v) air at 37 °C. After 24 h, the cells were treated with different concentrations of the tested compounds. 5-Fluorouracil was used as a positive control. After 24 h, cell viability was determined using the conventional MTT cell viability method as follows: the medium including MTT reagent (5 mg/mL) was added to each well, followed by 4 h incubation at 37 °C, and HCl-isopropanol solution was added to each well. After 4 h incubation in a dark place, the absorbance was measured at 570 nm using a Microplate Reader (MTP-310, Corona Electric, Hitachi, Japan).

4. Conclusion

In this contribution, thirteen compounds were isolated from the fruiting bodies of the fungus *Hericium erinaceus* and two of them are new natural products. The chemical structures of the new compounds were elucidated on the basis of spectroscopic studies. The anticancer potential of the isolated compounds were tested with an *in vitro* cytotoxic assay against six human cancer cell lines (SH-SY5Y, 1321N1, HCT-116, Caco-2, OVK18 and HeLa). Four compounds were found to suppress the growth of several human cancer cell lines.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.phytol.2019.04.017.

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